

Circulatory system

- a system of blood vessels with a pump and valves to ensure one-way flow of blood

single circulation (of fish):

- * blood passes through the heart only once in one complete circulation
- * blood pressure lower than mammals - too low for kidney function
- * blood oxygenated in gills
- * 2 chambers - 1 atria & 1 ventricle

double circulation (of mammals)

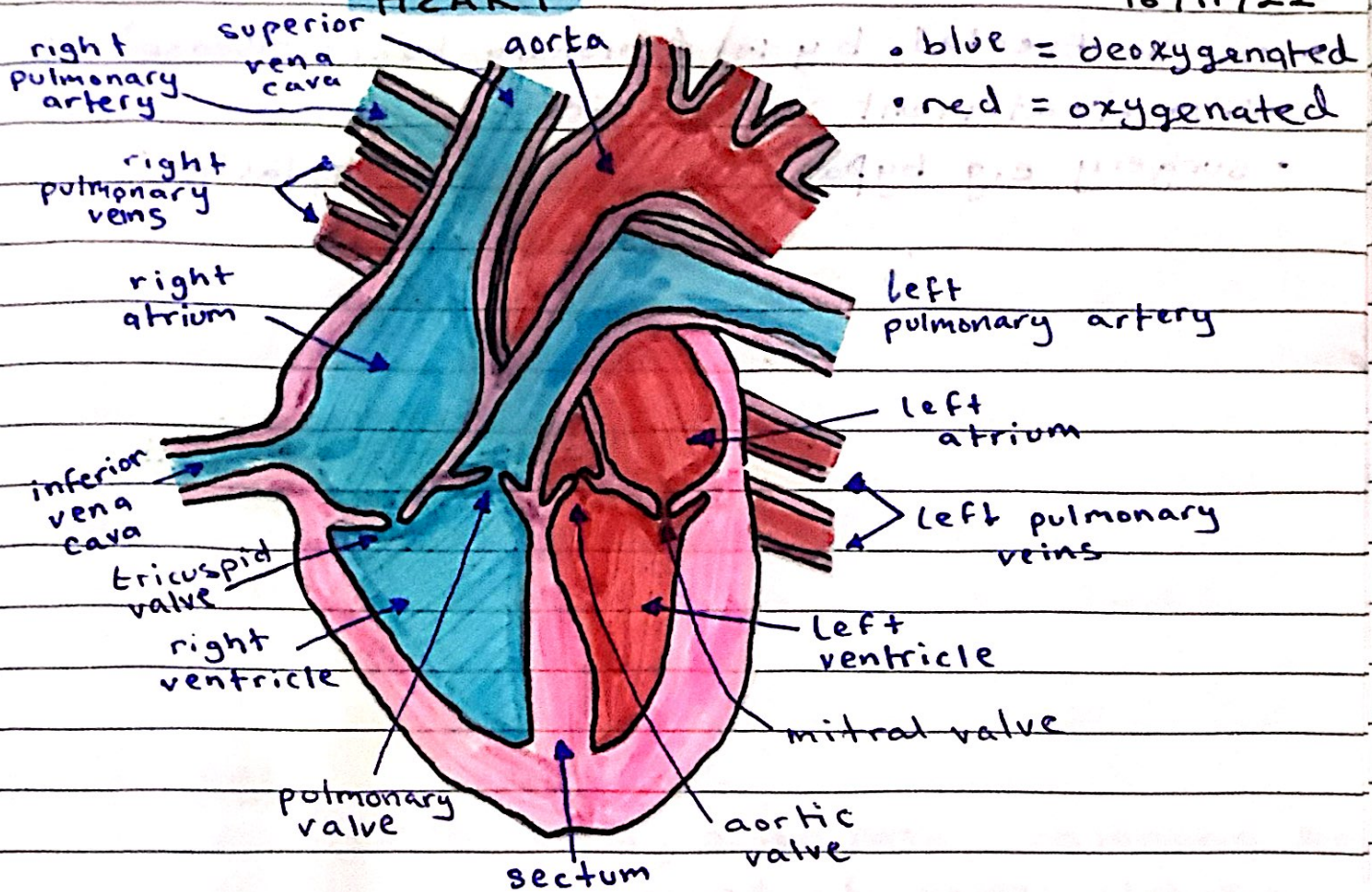
- * blood passes through the heart twice in one complete circulation
- * two separate pathways; for oxygenated and deoxygenated blood
- * pulmonary circulation = heart to lungs
- * systemic circulation = heart to rest of body

advantages of double circulation

- + separates oxygenated / deoxygenated blood so gas exchange is more efficient
- + higher pressure allows blood to be pumped around larger distances of the body
- + high pressure needed for kidney function and homeostasis
- + blood can be returned to heart faster; for exercise or to maintain a higher metabolic rate or control temperature in larger animals.

HEART

16/11/22



- * tricuspid & mitral valve = atrioventricular valve
- * pulmonary & aortic valve = semi lunar valve

comparing thickness:-

- left ventricle is thicker than the right because it has to pump blood all throughout the body while the right only pumps to the lungs.
- ventricles are thicker than the atria's because ventricles pump blood out and atria's only send the blood to the ventricles.

importance of septum:-

- forms a barrier between oxygenated / deoxygenated
- prevents them from mixing
- double circulation couldn't occur properly w/o septum.

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contraction of heart muscles:

* the blood enters the atria's, they contract to send the blood to the ventricles. Then the ventricles contract to send the blood to the lungs and body.

* atria contracts, ventricles relax
ventricles contract, atria relaxes

→ blood is pumped out of the heart through the arteries and returns to the heart through the veins.

heart rate

can be monitored by: ECG, pulse rate, listening to the sound of valves closing

- physical activity increases heart rate
- take pulse at rest, perform exercise, take pulse again. It will have increased
- muscles burn more oxygen, create waste product
so increased oxygen demand needs increased circulation

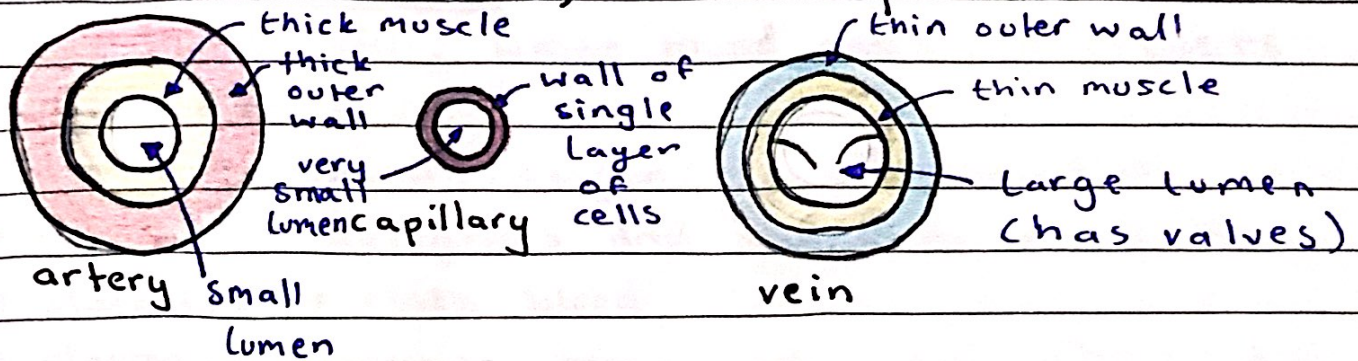
Coronary heart disease

* blockage of coronary arteries, vital tissues not getting oxygen

* risk factors: diet, stress, smoking, genetic predisposition, age and gender

* to prevent: - have good diet & lifestyle with low levels of cholesterol and saturated fatty acids
- do more exercise to strengthen heart muscle and reduce stress

Structure of arteries, veins & capillaries:



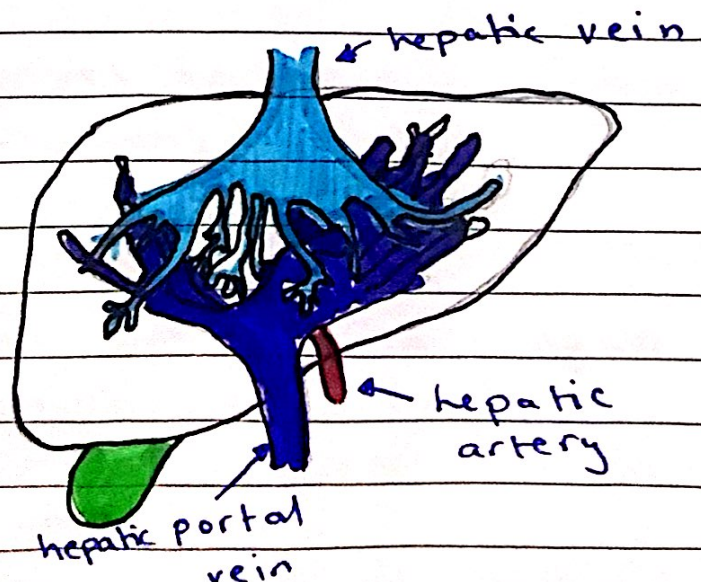
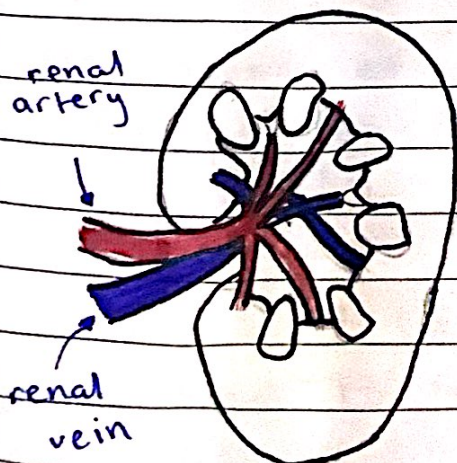
→ Lumen = area of cross section

• arteries are thick because they have to transport blood and withstand pressure while veins transport blood in lower pressure.

→ capillaries have very thin permeable walls which means substances can diffuse out very easily, useful substances are diffused out while waste products are absorbed.

Main blood vessels

- heart = vena cava, aorta, pulmonary vein/artery
- lungs = pulmonary vein/artery
- kidney = renal vein/artery
- liver = hepatic vein/artery and hepatic portal vein



components of blood

* red blood cells, white blood cells, platelets and plasma

→ RBC's → transport oxygen by haemoglobin

→ WBC's → phagocytosis and anti-body production

→ platelets → clots blood

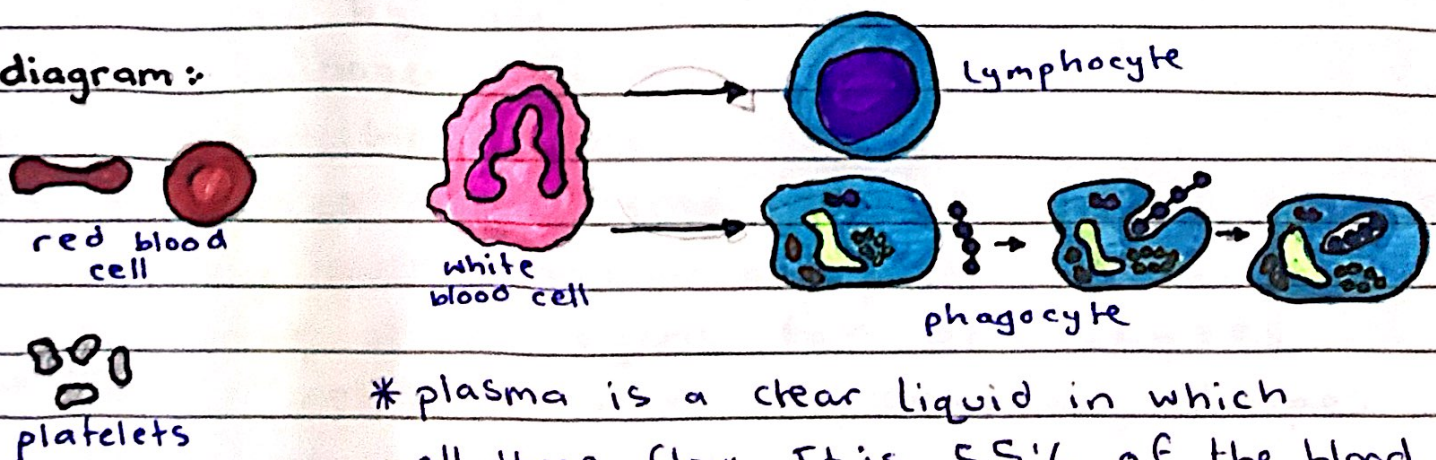
→ plasma → transports blood cells, ions, nutrients & waste

two types of WBC's:

Lymphocytes → help the body remember invaders and produce anti-body for them

phagocytes → engulf pathogens by phagocytosis

diagram:



* plasma is a clear liquid in which all these flow. It is 55% of the blood

clots blood at wounds/injuries to prevent blood loss and entry of pathogens

clotting blood is the conversion of fibrinogen to fibrin to form a mesh